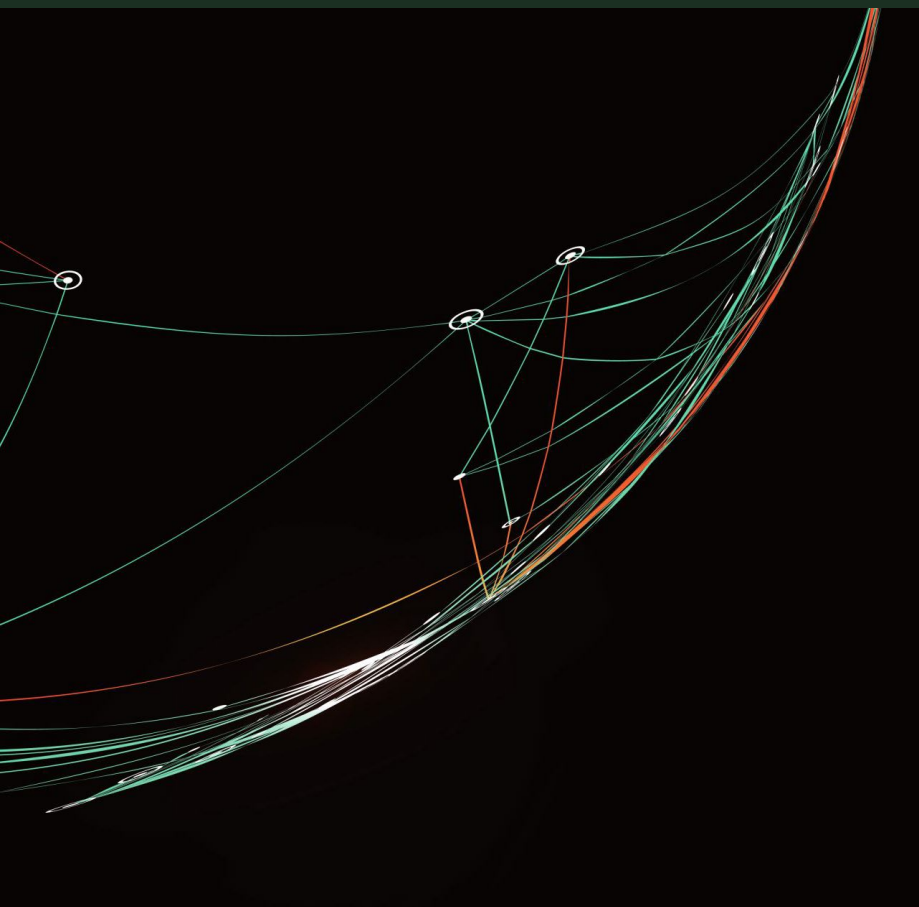




AI IN ACTION: FREE TOOLS FOR PUBLIC HEALTH DATA VISUALIZATION

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Johnson County Public Health

Presentation Outline



Data Analysis overview



Tools & Software explained



Demo

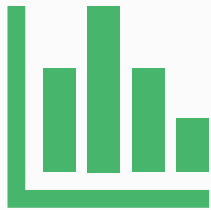


Resources

Data Analysis in Public Health

- Public health is data-rich, but time-limited
- Valuable insights often stay locked in spreadsheets
- Analysis and visualization is slow and resource-intensive
- AI tools can speed up routine analysis
- Goal: turn data into decisions—faster

AI in Data Analysis



**Help clean and
explore datasets**



**Create charts and
summaries quickly**



**Translate plain-
language questions
into results**



**Support faster
insight generation**

Overview of the tools



Python (Tool)

- **#1 Most popular programming language**
- **General-purpose language**
- **Easy readability**



Google Colab (Software)

- **Software that interacts with Python**
- **Part of the Google Ecosystem**
- **Easy accessibility**



Google Gemini (AI)

- **Generative AI model**
- **Generate and transform Python code**
- **Bounce ideas and help make inferences**

**Everything is Free*
to use!**

How the tools interact

Start:



Combines the data, AI, and Python into one place



Gemini

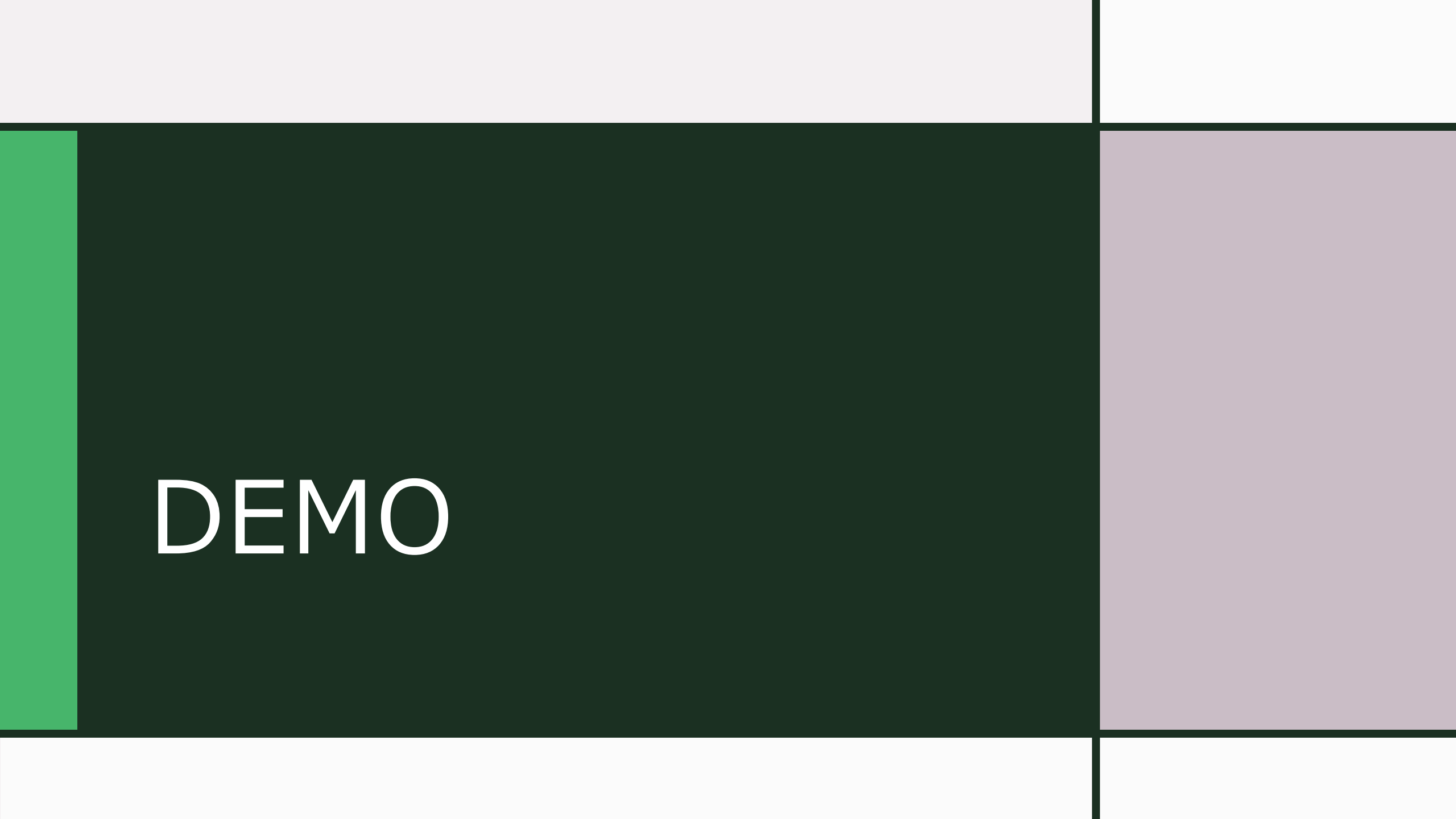
AI writes the Python code for the data to pass through based on the prompt given



End:



The Python code creates a chart based on the data passed through

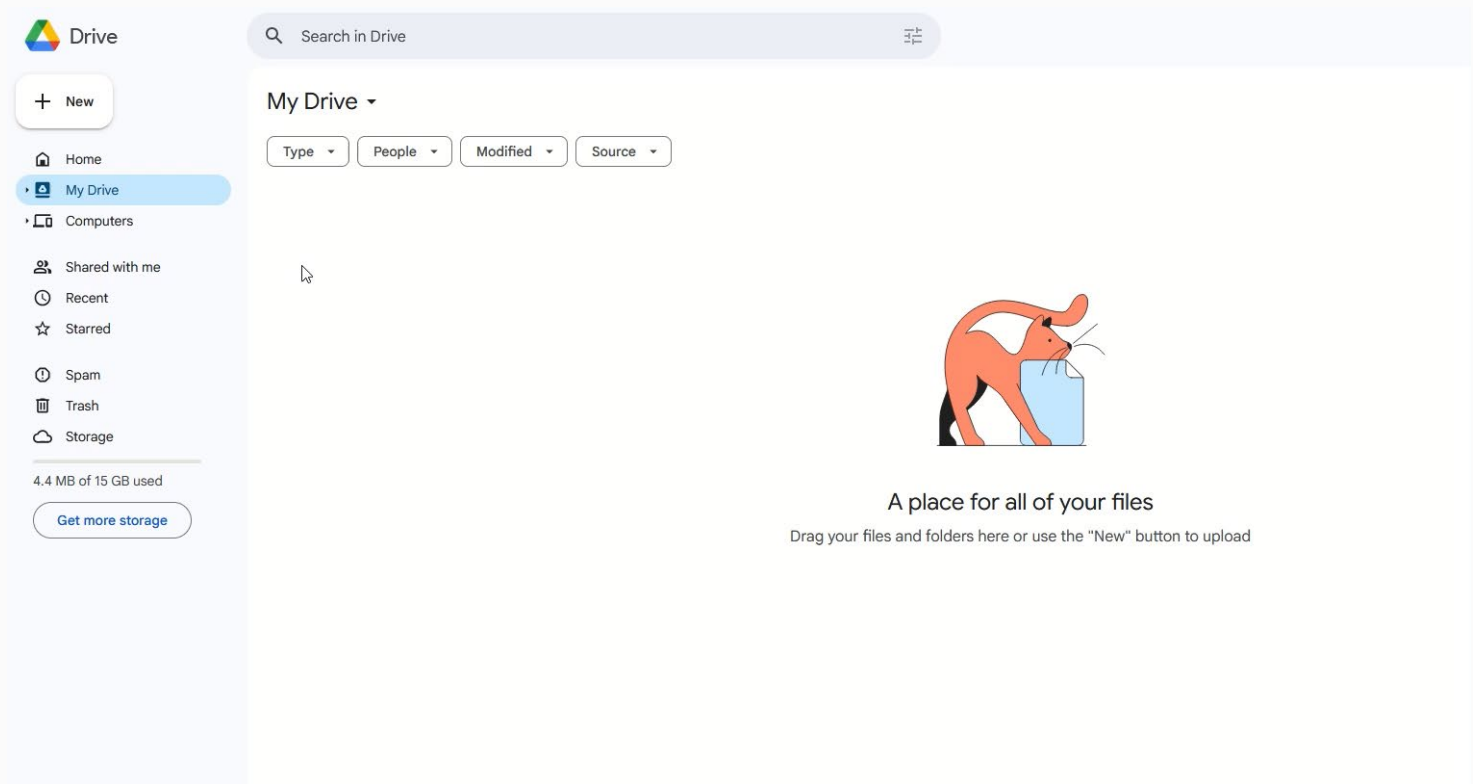


DEMO

Demo Objectives

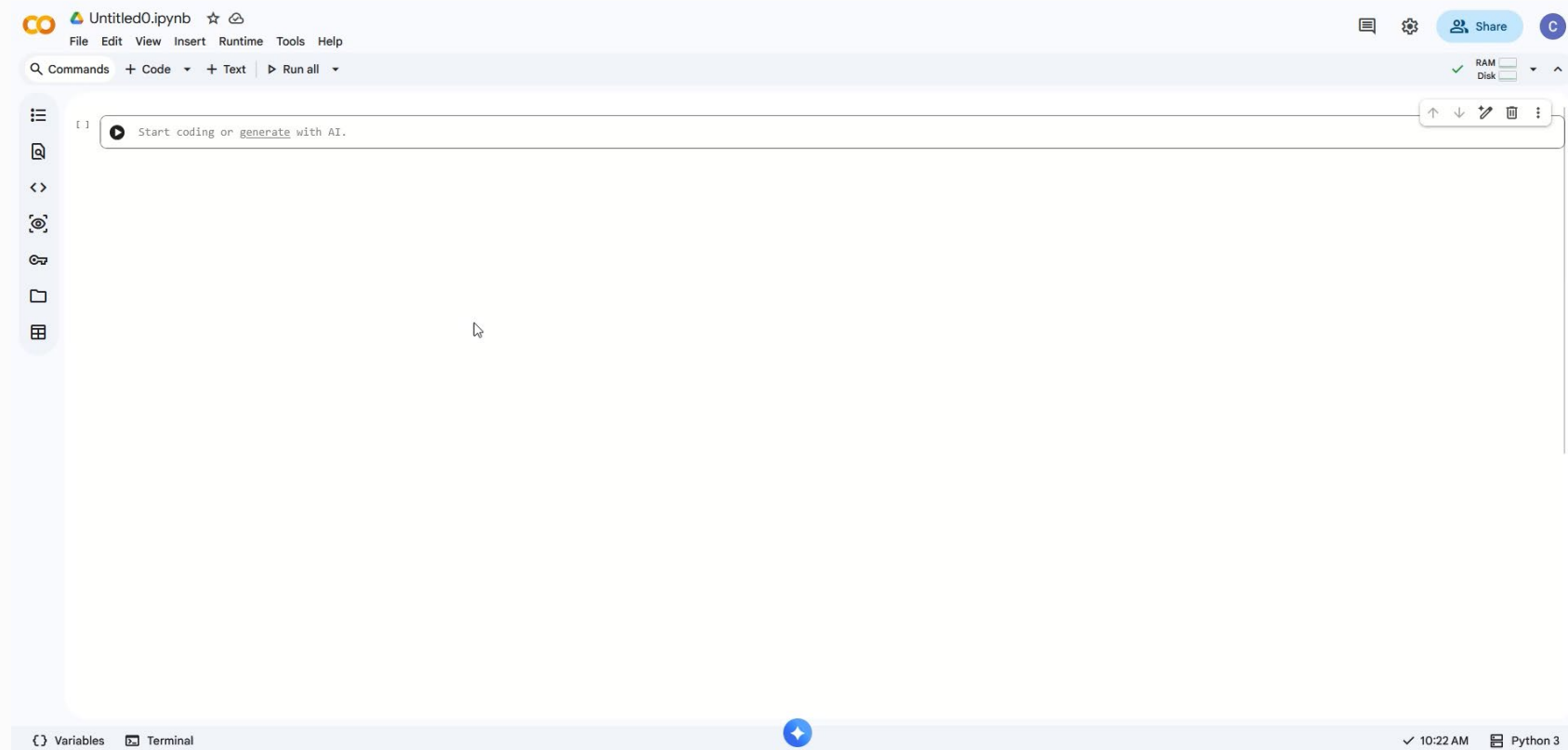
Goal: *Examine Smoking rates in Johnson County and compare against bordering counties.*

1. Download data from CDC Places (Already done)
2. Ask Gemini (AI) to create bar chart
3. Customize the bar chart to our liking with AI assistance



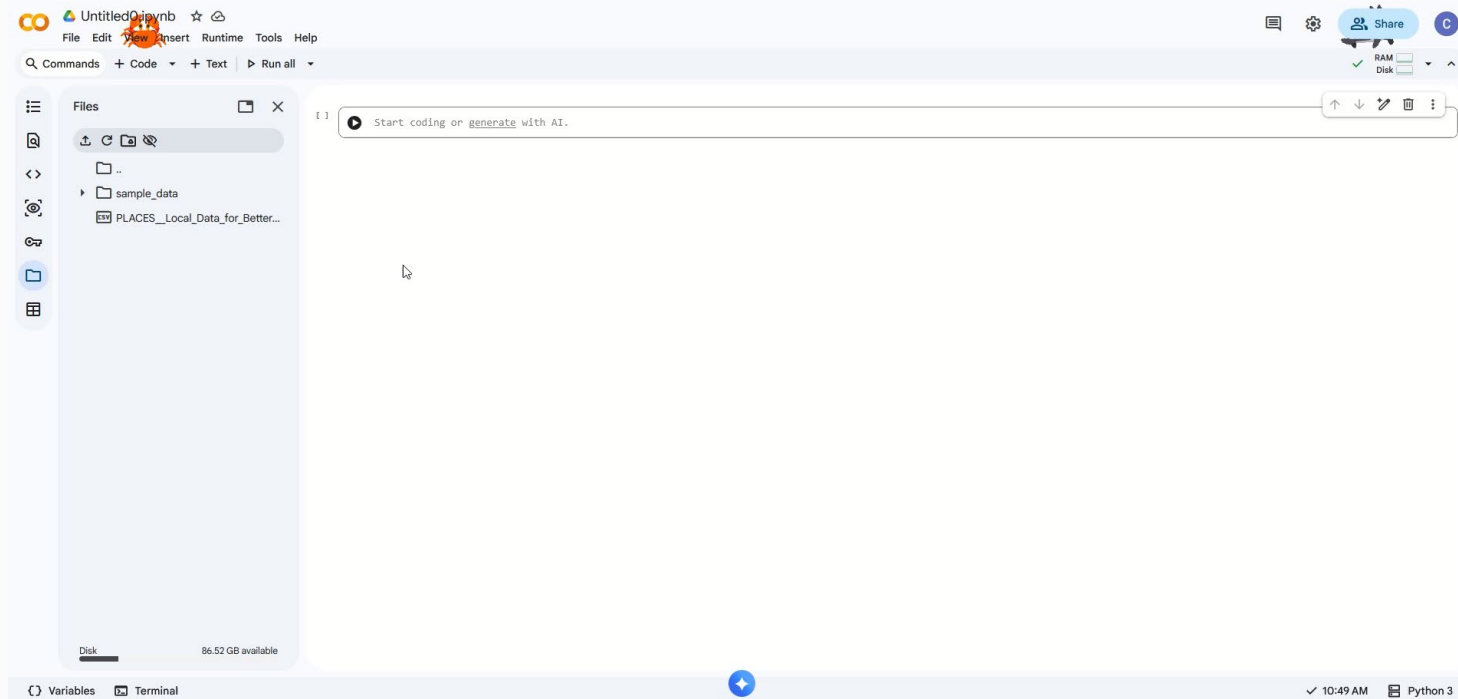
Getting Started

- Download Google Colab from the Google Marketplace
- Create a new Google Colab in your Google Drive



Setting Up

- Verify AI settings
 - Can either be enabled or disabled
- Enable fun easter eggs themes: Corgi, Kitty, and Crab mode (optional)
- Import your Data
 - Excel, CSV, TXT, XML, etc.



Exploring dataset

- Ensuring all systems can read the data
- See how the data is organized

The screenshot displays a Jupyter Notebook environment with a file explorer on the left and a command bar at the top. The notebook contains a single code cell with the following Python code:

```
import pandas as pd

df = pd.read_csv('/content/PLACES_Local_Data_for_Better_Health_County_Data_2025_release_20260130.csv')
display(df.head())
```

Below the code cell, the first five rows of the DataFrame are displayed as a table:

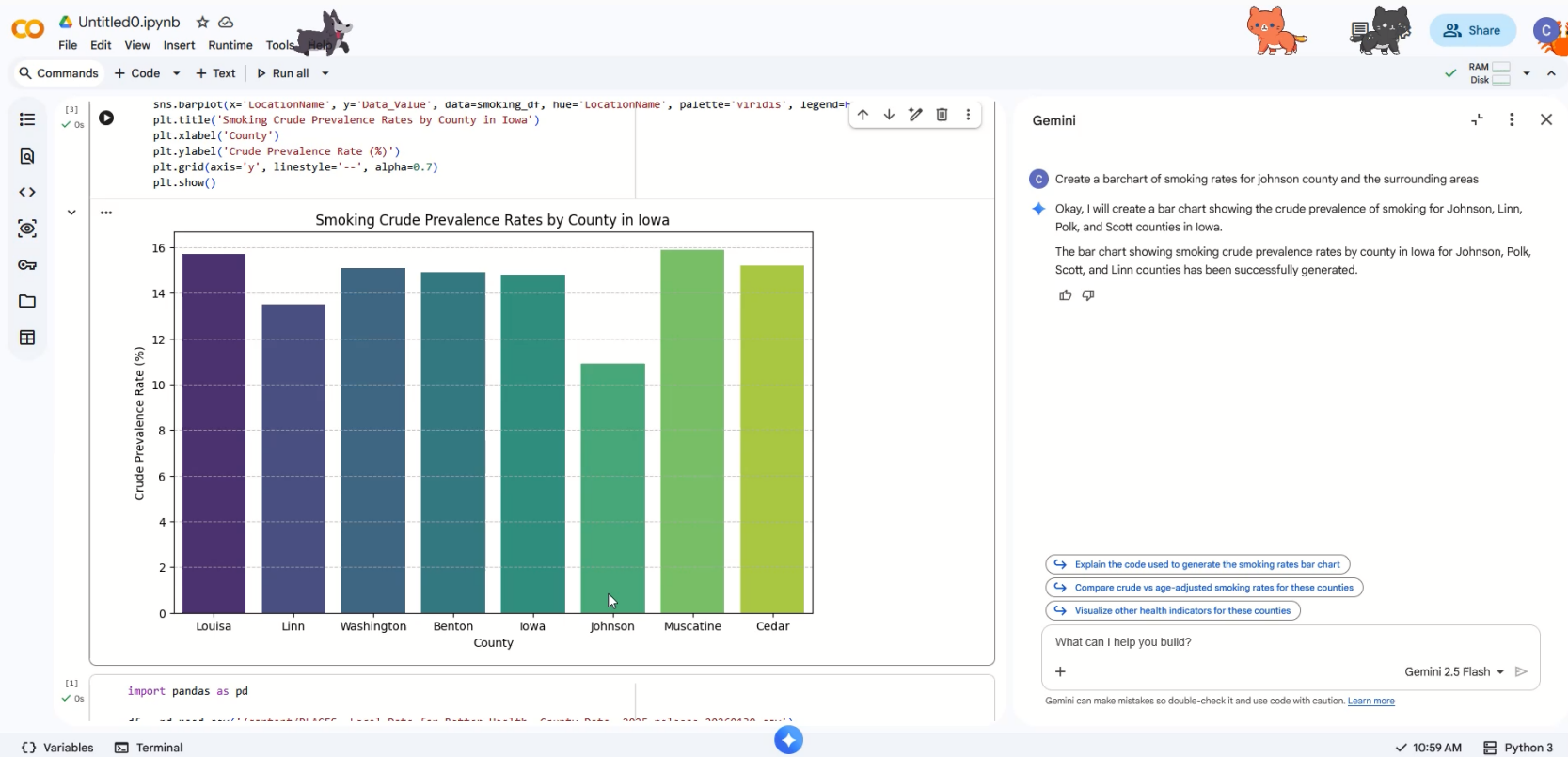
	Year	StateAbbr	StateDesc	LocationName	DataSource	Category	Measure	Data_Value_Unit	Data_Value_Type	Data_Value	...	Low_C
0	2023	IA	Iowa	Fremont	BRFSS	Health Outcomes	Depression among adults	%	Crude prevalence	20.1	...	
1	2023	IA	Iowa	Scott	BRFSS	Health Outcomes	Arthritis among adults	%	Crude prevalence	28.8	...	
2	2023	IA	Iowa	Tama	BRFSS	Health Outcomes	Obesity among adults	%	Crude prevalence	40.2	...	
3	2022	IA	Iowa	Shelby	BRFSS	Prevention	Mammography use among women aged 50-74 years	%	Crude prevalence	75.6	...	
4	2023	IA	Iowa	Calhoun	BRFSS	Disability	Hearing disability among adults	%	Crude prevalence	8.8	...	

Below the table, it indicates "5 rows x 22 columns".

To the right of the notebook is a Gemini chat window. The first message is: "Show me the first 5 rows of the data". The response is: "Sure, here's the code to load the data into a pandas DataFrame and display the first 5 rows: The first 5 rows of your dataset have been loaded and are displayed below. You can see columns such as 'Year', 'StateAbbr', 'StateDesc', 'LocationName', 'Category', 'Measure', 'Data_Value_Type', and 'Data_Value'." Below this, there are three suggested prompts: "Show me the column names and their data types", "Display summary statistics for the DataFrame df", and "Filter the data for a specific 'StateDesc'". The chat window also includes a text input field with the placeholder "What can I help you build?", a plus icon, and a dropdown menu set to "Gemini 2.5 Flash".

Creating a chart

- Ask the AI to generate a chart with the wanted metrics
- Can edit the Python code manually

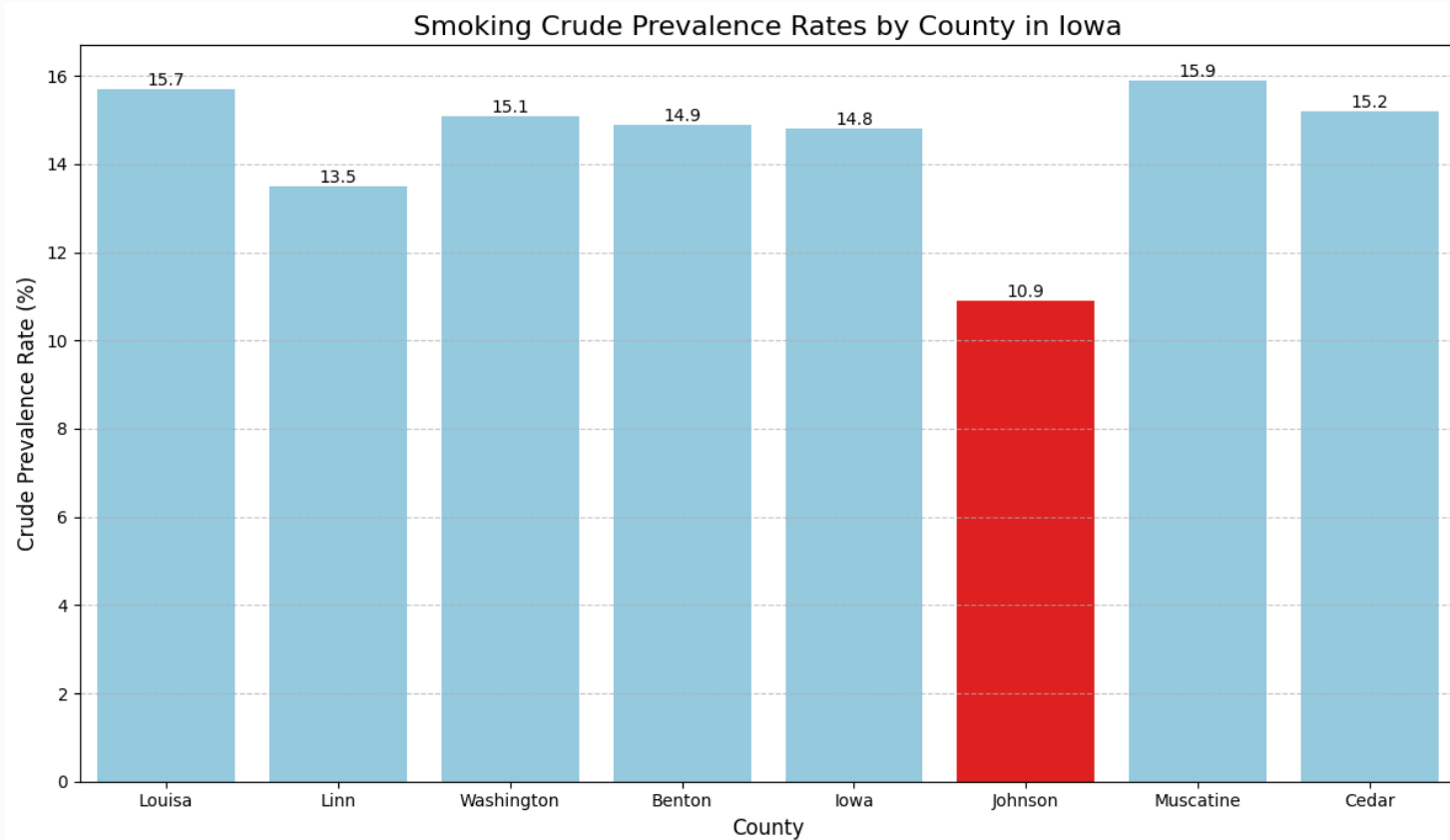


Customization

- Ask the AI to customize the generated chart further
- Can edit the Python code manually

Goal: *Examine Smoking rates in Johnson County and compare against bordering counties.*

Result:



Finished Result!

- Quick and clean bar chart
- Goal was to show the capabilities of AI and computer science for public health data visualization

Why learn Python?

- Built into widely used tools
- Broader benefits
 - Highly customizable
 - Automate tasks
 - Machine learning
 - Free and open source



Power BI



ArcGIS

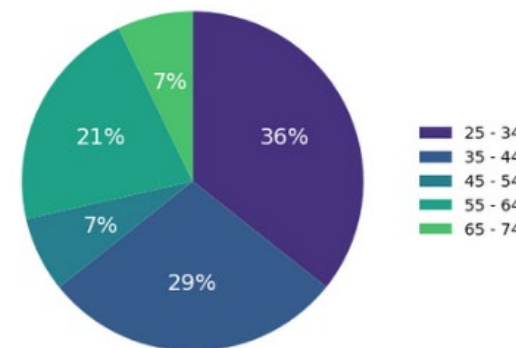


+ a b l e a u[®]



Examples of Charts made with Python

WHAT IS YOUR AGE? (N=14)



Average Hourly EMS Dispatches per Month where Narcan was Administered

Narcan administrations may include cases beyond suspected opioid overdose

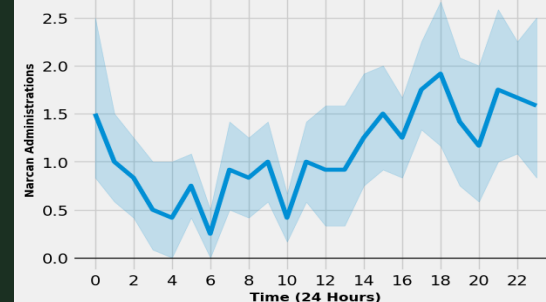
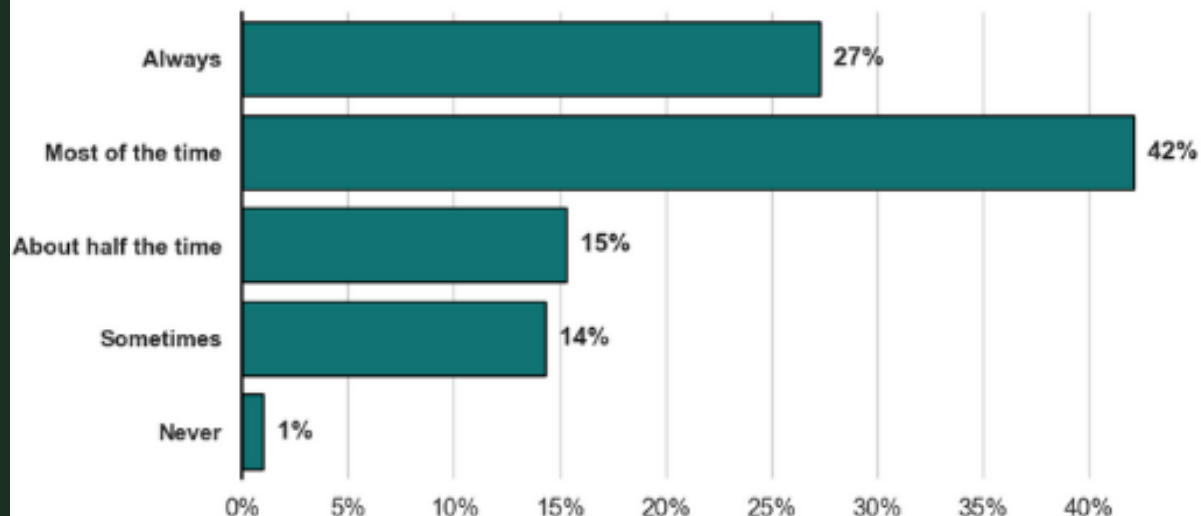
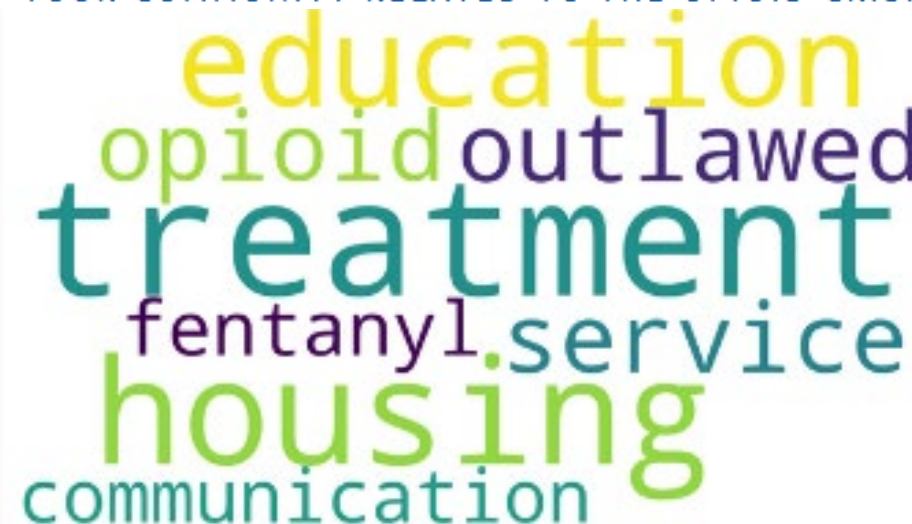


FIGURE QV.3. DIETARY NEEDS BEING MET



WHAT ARE THE GREATEST UNMET NEEDS THAT EXIST IN YOUR COMMUNITY RELATED TO THE OPIOID CRISIS?



Where to Learn Python

- Book: *Python for Data Analysis, 3E* by Wes McKinney
- Website: <https://www.kaggle.com/learn>

Check out Johnson County Public Health

- <https://healthyjoco-johnsoncounty.hub.arcgis.com/>
- <https://johnsoncountyiowa.gov/department-of-public-health>

Resources

CDC PLACES Data Source:
<https://data.cdc.gov/d/swc5-untb>